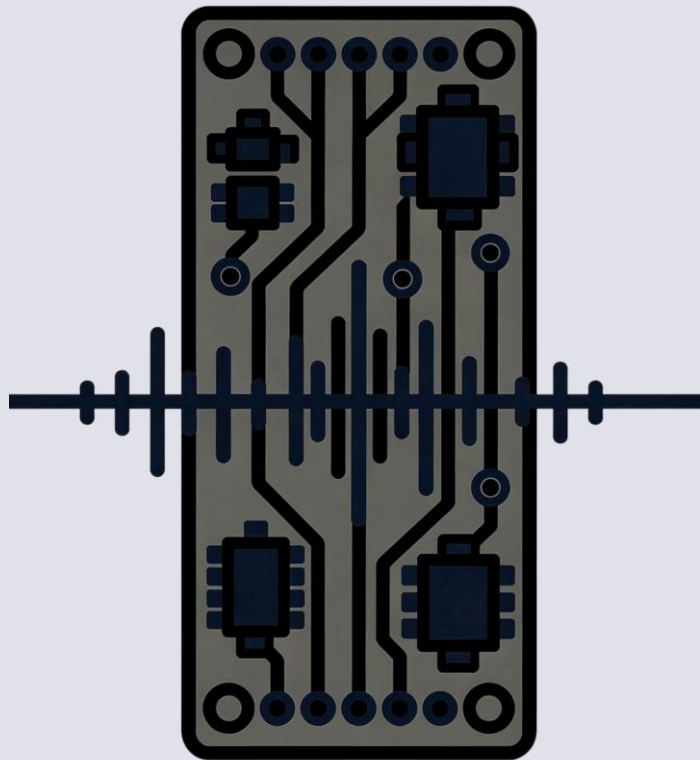


Bounding the Unpredictable

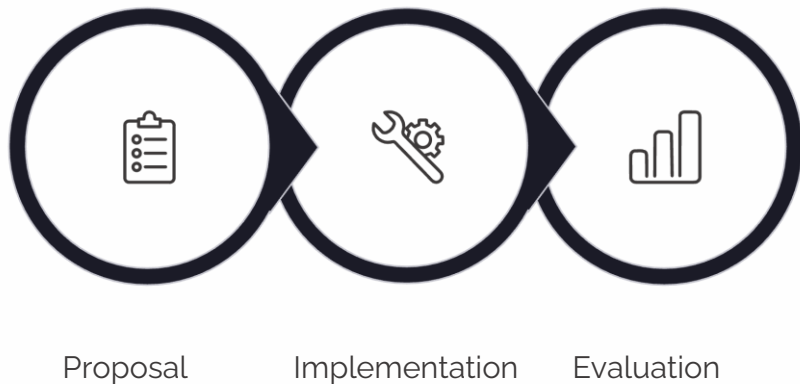
Deterministic Audio Latency on Linux SoCs

Implementation Progress and Experimental Evaluation

Goal: Establish a reproducible simulation-based framework to bound worst-case audio latency on Linux SoCs



What Was Proposed?



Research Gap

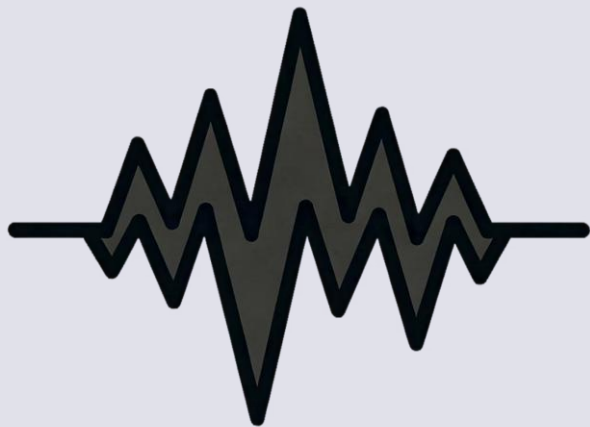
No reproducible simulation-based framework for end-to-end audio latency analysis on embedded Linux

Dual Methodology

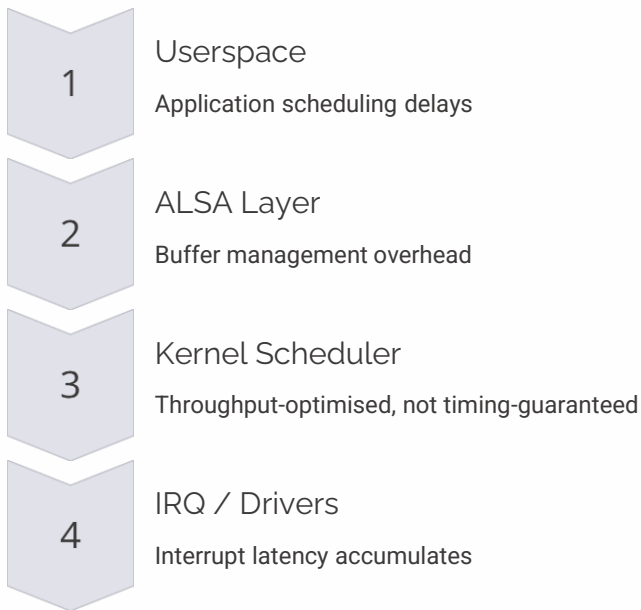
Empirical latency measurements + predictive latency budget model

Key Comparison

Standard Linux kernel vs PREEMPT_RT under controlled stress profiles

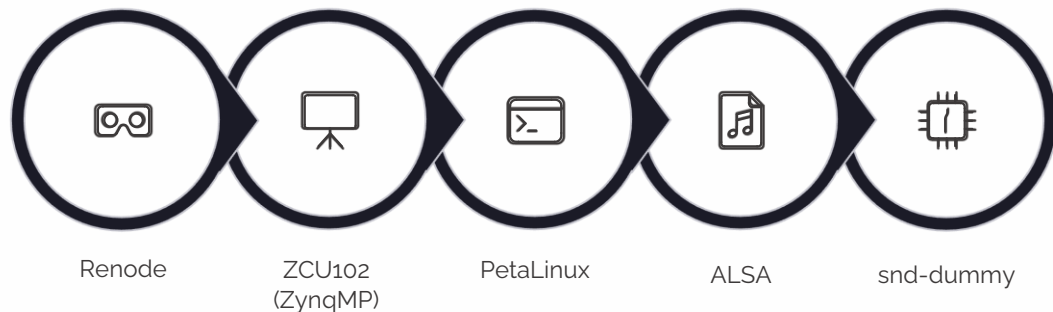


Why Audio Determinism Is Difficult



Late audio frames cause glitches or deadline misses — latency accumulates across every layer.

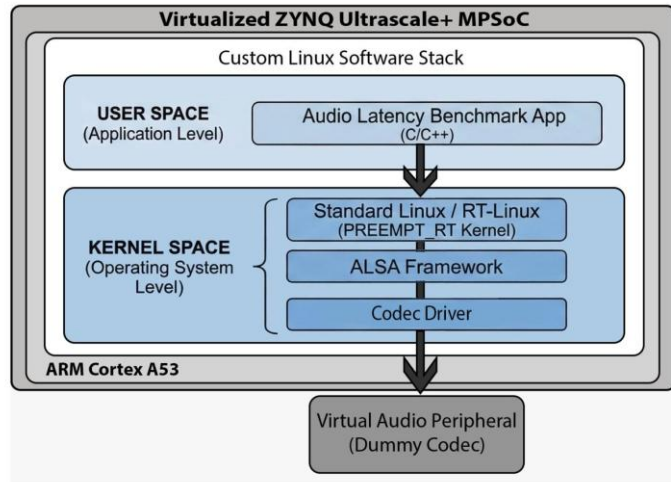
Implemented Platform



XSA (Xilinx Support Archive) is the hardware description exported from Vivado that tells PetaLinux what hardware design exists on the FPGA/SoC, including processors, memory, clocks, and peripherals.

BSP (Board Support Package) is the software configuration package that provides the board-specific boot files, device tree, kernel settings, and hardware support needed to build and run PetaLinux on a target board.

snd-dummy enables ALSA-level audio timing experiments without requiring a fully modelled physical audio peripheral.



Renode + PetaLinux: Integration Challenges

Challenge

ZynqMP platform / device-tree alignment

Kernel modules and ALSA components missing

Hardware simulation problems

Rootfs size and simulation overhead

Resolution

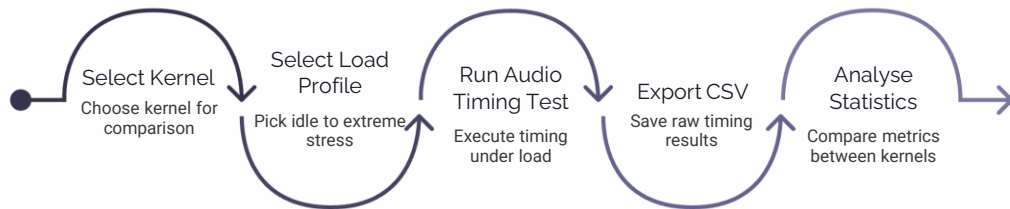
Matched Renode platform script to PetaLinux DTB

Configured PetaLinux build with required modules

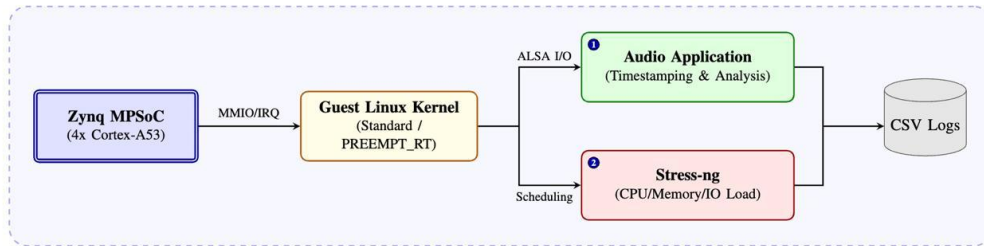
Problematic hw / operations disabled

Stripped minimal rootfs for experiment environment

Experimental Design



Renode Simulation Environment (Deterministic Virtual Time)



Load Profiles

Idle
Baseline, no stress

Light
Moderate CPU load

Heavy
High CPU + memory pressure

Extreme
Near-saturation stress

Configuration	Period (-p)	Buffer (-b)	Start Thr. (-t)	Theoretical Latency @48 kHz	Purpose
Profile A (Safe)	128	512	128	~2.66 ms	Reference safe-limit behaviour.
Profile B (Low)	64	256	64	~1.33 ms	Aggressive low-latency boundary condition.
Profile C (Ultra-Low)	32	128	32	~0.66 ms	Expose the system breaking point under tight buffering.

Measurement Methodology

Mean Latency

Baseline performance indicator across repeated measurements.

Max / P99 Latency

Tail behaviour – where determinism fails under stress.

Jitter / Spread

Variance across repeated measurements; reveals timing instability.

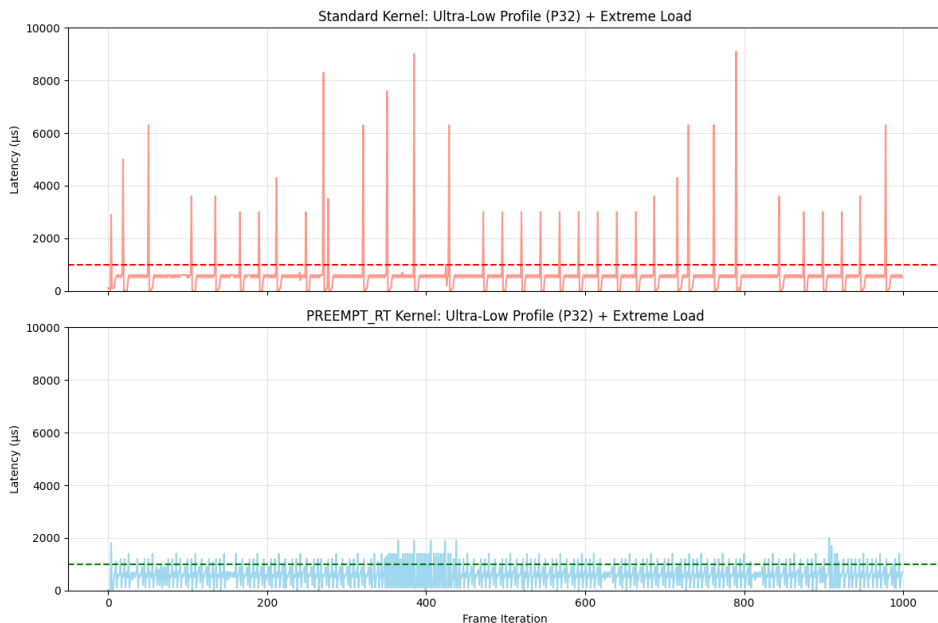
XRUN Frequency

Buffer underrun/overflow event count; indicates timing failures in the audio pipeline.



Deterministic audio should not be evaluated only by average latency. Tail latency, jitter, and XRUN frequency reveal whether the system can consistently meet timing deadlines.

Results: Determinism Under Extreme Load

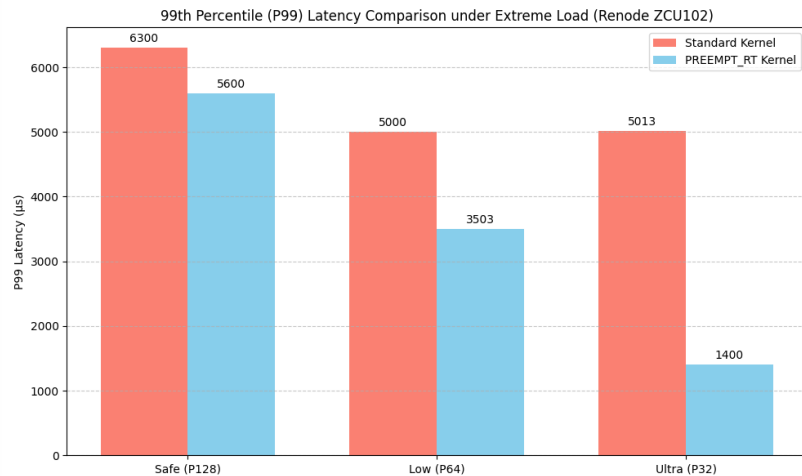
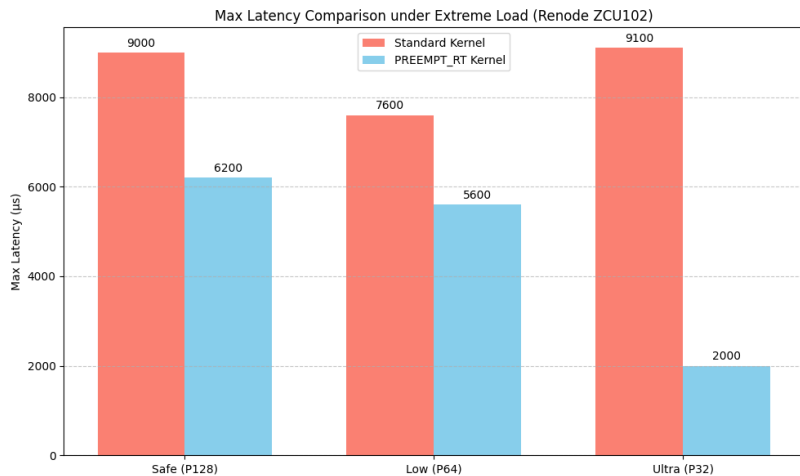


Key Observations

- Standard Linux exhibits significant latency spikes under extreme contention.
- PREEMPT_RT maintains tighter latency bounds and improves temporal predictability.
- Worst-case behavior becomes increasingly important for deterministic audio systems.
- Real-time scheduling improves robustness against CPU and memory pressure.
- PREEMPT_RT provides more stable timing behavior under worst-case operating conditions.

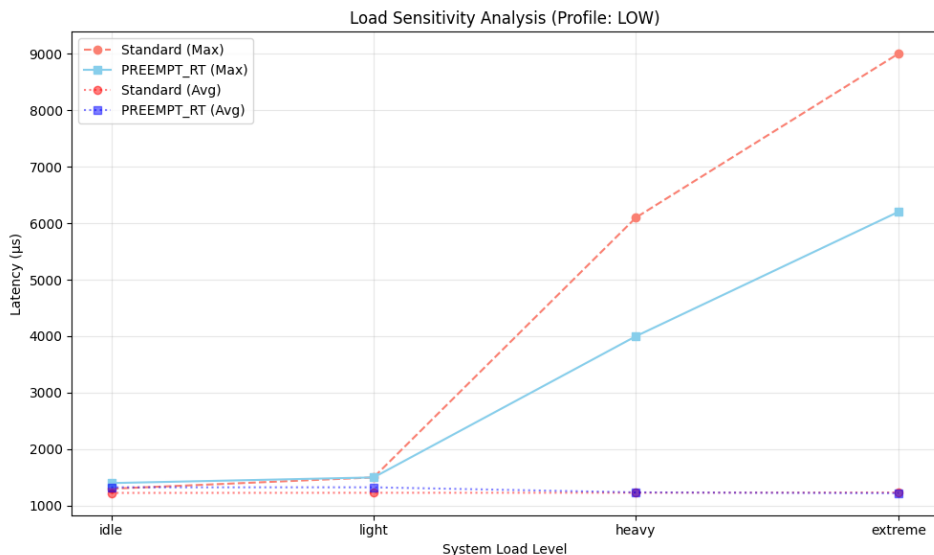
RESULTS

Results: Latency Characteristics: Maximum vs P99



- Key Observations:**
- Both maximum and P99 latency metrics show that PREEMPT_RT performs better than Standard Kernel.
 - As load increases, the difference grows.

Results: Load Sensitivity Analysis



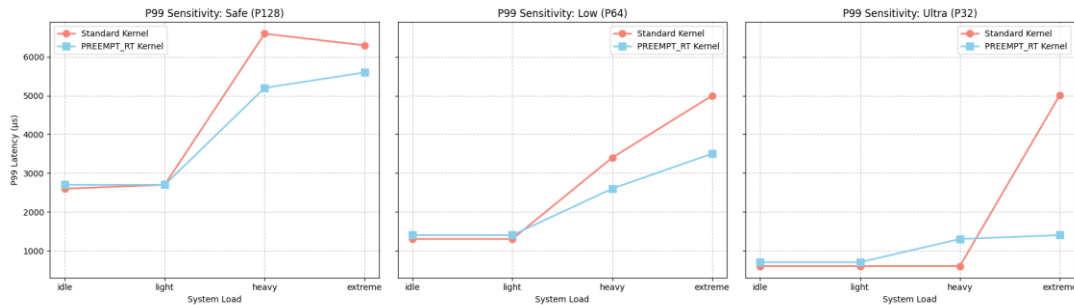
Key Observations

- Standard Linux exhibits greater performance degradation as load intensity increases.
- PREEMPT_RT maintains more stable latency characteristics under contention.
- Both kernels exhibit the same average latency under every load condition.
- Real-time scheduling improves robustness against CPU and memory pressure.

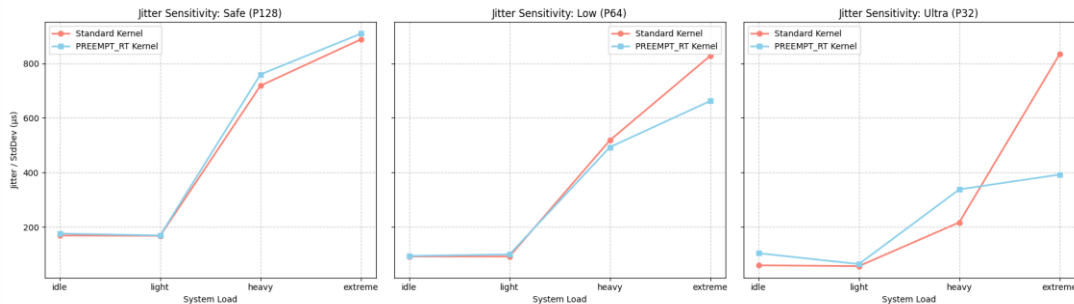
RESULTS

Results: Load Sensitivity Analysis (P99 and Jitter)

P99 Latency Sensitivity to System Load (Rende ZCU102)



Jitter (Standard Deviation) Sensitivity to System Load



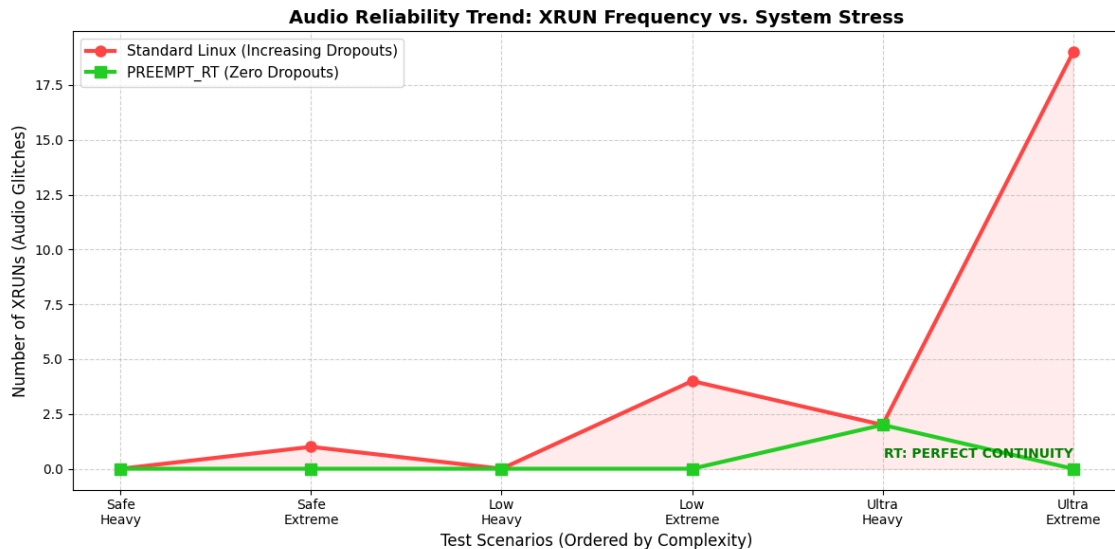
Key Observations

- PREEMPT_RT improves timing predictability under heavier contention.
- Jitter behavior may vary across load levels due to scheduling dynamics.
- Tail latency and timing stability jointly determine deterministic performance.

RESULTS

Results: XRUN Analysis

XRUN (Buffer Underrun) means the CPU failed to provide audio data in time, resulting in an audible 'pop' or silence.



Key Observations

- XRUN events indicate timing instability and deadline violations.
- System load directly impacts audio pipeline reliability.
- PREEMPT_RT improves robustness under stressed operating conditions.

Proposal vs Implementation

Proposal Item	Status	Evidence / Comment
Renode simulation framework	✓ Achieved	Platform booted and running
PetaLinux boot and configuration	✓ Achieved	ZynqMP image configured
ALSA audio pipeline execution	✓ Achieved	Via snd-dummy driver
Standard vs PREEMPT_RT comparison	✓ Achieved	CSV data collected for both
Stress-profile based testing	✓ Achieved	4 load profiles executed



What Remains To Be Done

01

Extend Audio Device Model

Move beyond snd-dummy toward a more realistic audio device

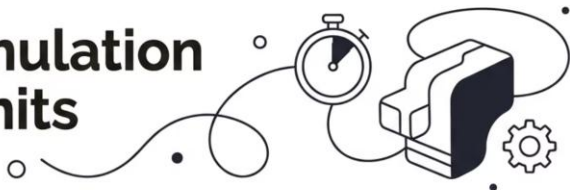
02

Finalise Latency Budget Model

Compare predictive model against empirical CSV measurements

Discussion and Limitations

Simulation Limits



Simulation

Renode gives reproducibility but not full physical timing fidelity

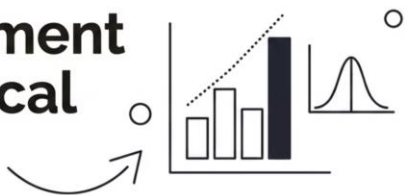
Audio Device Abstraction



snd-dummy

Useful for ALSA/software-layer testing; does not capture all real hardware delays

Measurement & Statistical Limits



PREEMPT_RT

Improves scheduling; determinism still depends on drivers, IRQs, buffering, and userspace

CONCLUSION

Key Takeaways



Working Setup

Renode + PetaLinux + ALSA and
snd-dummy experimental framework
successfully implemented



Comparison Done

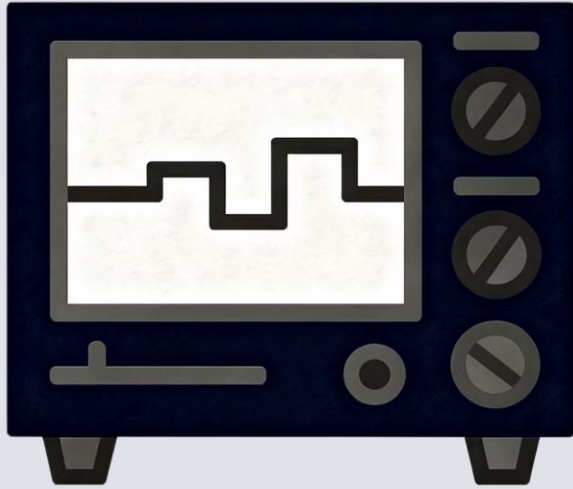
Standard Linux vs PREEMPT_RT
measured across four controlled
load profiles



Next Step

Connect empirical results to the
predictive latency budget.

"Determinism is not achieved by measuring average speed; it is evaluated by **bounding worst-case timing behaviour**."



Thank You

CMP720 - Embedded Systems Design